

A multi-channel iPSC-cardiomyocyte repolarization model does not beat a hERG-only model on CiPA risk classification

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Abstract

A multi-channel model of the human induced-pluripotent-stem-cell cardiomyocyte (iPSC-CM) action potential must do more than a hERG-only model before it can be adopted in a cardiotoxicity screen. We test one such model, the Kernik-2019 iPSC-CM model run with a three-current Hill block, against that bar on two endpoints and two measured datasets. On the three-class CiPA risk label the model scores 17 of 28 drugs, but that score is fragile: sweeping the supplied free exposure gives 16 of 28, with diltiazem the only change. A hERG-only model, scored on the same 12/16 fit-and-freeze split, scores 18 of 28 on the static hERG input and 19 of 28 on the dynamic input. The hERG-only model scores higher. On the continuous repolarization readout the model does better: against Blinova et al. (2018), with a Fridericia rate correction applied to both sides, the model reaches a Spearman rank correlation of 0.70 over 112 matched drug-dose points, against 0.62 for the hERG-only model, and its direction agreement on the measurable points is 82 of 92 (89%) against 72 of 92 (78%) for both the hERG-only model and a trivial always-prolong rule. Against Lee et al. (2025), the model recovers the measured FPDc rank at 0.67 over the 75 non-quiescent drug-dose cells, but that correlation is dose-dependent: 0.55 at $1 \times C_{\max}$ and 0.84 at $10 \times$. Direction agrees on 9 of 10 anchored drugs, against 6 of 10 for the hERG-only model. The added channels capture the shortening direction of calcium-channel blockers, and they raise the rank correlation. They do not improve the three-class risk label. We also report four negative or null results: the population confidence signal is overconfident under both population models, adding I_{Ks} is null, adding I_{NaL} degrades accuracy and moves nine drugs, and the model’s own mechanism recovery fails on most single-current and current-pair tests. The comparison against the hERG-only model, the full failure accounting, and the code and data are published so the result can be reproduced.

Keywords: iPSC-cardiomyocyte, action potential, cardiotoxicity, validation, CiPA

1. Introduction

In-silico models of the human ventricular action potential are proposed as an early cardiotoxicity screen. A single strong current, the rapid delayed rectifier I_{Kr} (hERG), dominates the QT-prolongation signal. A multi-channel model therefore has to clear a specific bar: it

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must do more than a hERG-only model before it can be adopted in a safety package. This paper tests that bar.

The model is the Kernik-2019 human iPSC-CM action-potential model [1]. It is an iPSC-CM model, the substrate CiPA recommends. We run it against two independent measured datasets: Blinova et al. (2018) [2], the CiPA multi-site myocyte validation study, and Lee et al. (2025) [3], a 28-drug iPSC-CM screen.

We ask two questions. First, on the three-class CiPA risk label, does the multi-channel model beat a hERG-only model on the same fit-and-freeze split? Second, on the continuous repolarization duration, does it reproduce the direction and rank of the measured response better than the hERG-only model? The two answers differ, and that difference is the finding.

2. Methods

Model and solver. The substrate is the Kernik-2019 iPSC-CM CellML model [1], run through Myokit with the CVODE (SUNDIALS) adaptive solver. The model is autonomous: it carries the Kernik I_f pacemaker current and no external stimulus, so it beats spontaneously at about 61 beats per minute at baseline. Drug-free baseline values are $APD90 = 413.31$ ms, $MDP = -75.60$ mV, cycle length = 982.9 ms, and rate = 61.0 bpm.

Simulation protocol. Every condition starts from the CellML initial state. We pre-pace for 10,000 ms to settle the limit cycle, then log 8,000 ms at a 0.1 ms logging interval. An upstroke is detected where dV/dt crosses a fixed threshold. Cycle length is the mean interval between successive peak-voltage times, and $APD90$ is the mean interval from upstroke to 90% repolarization over the detected beats. A condition with fewer than two clean beats is recorded as missing. The 8,000 ms window is set so that even a slow-beating, strongly blocked cell holds at least three upstrokes.

Drug block. A compound is applied as fractional Hill block of the conductances g_{Kr} , g_{CaL} , and g_{Na} (peak). The blocked fraction at concentration c is $1/(1 + (c/IC50)^h)$, with the reported $IC50$ and Hill coefficient h . Exposures are the drug’s free maximal concentration (C_{max}) multiplied by 1, 2, 3, and 4. When a drug has no reference $IC50$ for a current, that current is not blocked. The hERG input is the static Li-2017 fit distributed with the CiPA reference set (FDA/CiPA Li2017_IC50.csv, GPL-3.0); we also score a dynamic hERG column as a cross-check. The I_{CaL} and I_{Na} (peak) inputs come from the same reference set.

Risk label. The risk score is the worst-case fractional $APD90$ prolongation across the 1–4× exposure grid. A cell that fails to repolarize is a collapse. The simulation marks a collapse either by repolarization failure or by sustained depolarization above -20 mV for more than 50% of the trace. A collapse carries no duration, so it is recorded at a fixed ceiling of twice the baseline prolongation. The ceiling is 826.6 ms of prolongation (an $APD90$ of 1,240 ms). It is a plotting convention and carries no predictive claim. Two cut points on the continuous score define the CiPA Low, Intermediate, and High classes. The cuts are fit on the 12 training drugs and then frozen; the 16 validation drugs are scored blind.

Metrics. We report the Spearman rank correlation between model APD90 prolongation and measured FPD prolongation over three sets: all paired points, points with a completed beat only, and points measurable on both sides (absolute response above 5 ms). The 112 Blinova points are 28 drugs at four doses, and both variables rise with dose inside a drug, so the pooled correlation counts each drug four times. The pooled number is therefore inflated. We also report the per-dose correlation (one value per drug, $n = 28$ at each dose) and the per-drug correlation (one point per drug, from the mean over doses and the maximum measured response). Direction agreement is the sign match between model and measurement. We test it against a base rate: on the measurable set most points are positive, so an always-prolong rule scores 72 of 92 (78%), and we test the model’s excess over that base rate with a one-sided binomial test. Class balance is reported because the operating point issues few or no Low calls.

Rate correction. Blinova et al. and Lee et al. report Fridericia-corrected field potential duration (FPDc). The model is a spontaneously-beating cell whose cycle length changes with drug, so its uncorrected APD90 prolongation is not the quantity the measurement reports. We apply the same correction to the model, $\text{APD90c} = \text{APD90}/(\text{CL}_s)^{1/3}$ with cycle length in seconds, and recompute every Blinova statistic on the corrected delta. We report both the uncorrected and the corrected rank, and the corrected value is the primary result.

Data. The Blinova et al. (2018) analysis uses the released per-well Excel file. The measured response at each drug and dose is the median of the released per-well $\Delta\Delta\text{FPDc}$ across sites, cell types, and platforms, following the study’s own multi-site aggregation. The contributing record count is uneven: it ranges from 1 (bepridil at $4\times$) to 81. The Lee et al. (2025) comparison uses the drug directions and FPDc values stated in the main text. For the full panel it uses the FPDc heatmap in Figure 3 of that paper. The supplement reports per-drug percentage change in FPD as image-only tables (Tables S2 to S4), which we did not digitize. Figure 3 gives FPDc at four C_{max} ratios ($0.1\times$, $1\times$, $10\times$, $100\times$). We digitize the figure into a signed ordinal in $[-4, +4]$ (`lee_fig3_extract.py`) and rank the model against it. The ordinal is coarse, so we report the tie count at each dose and treat the high-dose cells with caution.

3. Results

3.1. The hERG-only model comparison on the CiPA risk label

The hERG-only model scores higher. Table 1 gives the fit-and-freeze result for the multi-channel model and for a hERG-only model on the same data, split, and protocol. The hERG-only model reaches 18 of 28 on the static hERG input and 19 of 28 on the dynamic input; the multi-channel model reaches 17 of 28. One compound separates them on the static column, and two on the dynamic column.

The model’s 17th correct call is fragile. The score of diltiazem, the only Low drug the model calls correctly, sits at -0.104 against a low cut of -0.1044 , a margin of order 10^{-3} , and that cut is itself fit on diltiazem’s own score. A sensitivity sweep over the supplied free C_{max} gives 16 of 28 at the nominal exposure, with diltiazem the sole difference. We treat 16 as the operating figure.

Table 1: CiPA three-class benchmark, 12 training drugs with cuts fit and frozen, 16 validation drugs scored blind. The hERG-only model blocks g_{Kr} alone, on the static or the dynamic hERG IC50 column. The multi-channel model adds g_{CaL} and g_{Na} (peak). The hERG-only model is the bar the Introduction sets.

Classifier	Train (12)	Blind (16)	All 28	Fitted cuts (lo / hi)
Multi-channel ($I_{Kr} + I_{CaL} + I_{Na}$)	8	9	17	−0.1044 / 0.3491
hERG-only model, static	8	10	18	−2.0342 / −0.5893
hERG-only model, dynamic	8	11	19	−1.7312 / −0.6569

The head-to-head on individual drugs does not favor the model. The model is right and the hERG-only model is wrong on two drugs, astemizole and risperidone. The hERG-only model is right and the model is wrong on three, loratadine, nifedipine, and nitrendipine. Verapamil is wrong for both. Astemizole is the strongest single case for a multi-channel read: it was withdrawn over QT prolongation, but its hERG IC50 (33.3 nM) is 128 times its free C_{max} (0.26 nM), so the hERG-only model calls it Low while the model calls it Intermediate, correctly. That one case does not offset the aggregate.

The operating point also has a class-balance defect. At the nominal exposure the model issues 21 Intermediate calls and 7 High calls, and no Low calls at all. Nine drugs are Low. In practice the three-class read is two-class, and it is wrong on every Low drug.

3.2. The hERG-only model comparison on the continuous repolarization readout

The result on the continuous readout is different. Against Blinova et al. (2018), with the Fridericia correction applied to both sides, the multi-channel model reaches a Spearman rank correlation of 0.70 over 112 matched points, against 0.62 for the static hERG-only model and 0.57 for the dynamic hERG-only model. Excluding collapse points the model reaches 0.67, against 0.54 for the hERG-only model. The model scores higher at every dose and on the per-drug correlation (Table 2).

We compare direction agreement with a trivial always-prolong rule. On the points measurable on the measured side, 92 of the 112, the model agrees with the measured sign on 82 (89%). Both hERG-only models agree on 72 of 92 (78%). That 78% is the always-prolong base rate: 72 of the 92 measurable medians are positive, and the hERG-only model always prolongs. The model’s 82 of 92 is 10 points above the base rate (one-sided binomial $p = 0.0053$). The hERG-only model has no excess to report, because it cannot go negative. The model produces a negative direction on only four drugs: diltiazem, loratadine, nifedipine, and nitrendipine.

Direction agreement decays with dose. It is perfect at the clinical exposure and falls across the grid: 18 of 18 at $1\times$, 22 of 23 at $2\times$, 23 of 26 at $3\times$, and 19 of 25 at $4\times$. The failure mode at high dose is the model over-prolonging. The measured response saturates or reverses while the model keeps the block increasing.

The 5 ms filter removes points whose measured response is small, so it is a measurement-side filter. All 20 excluded points fail the measured-side cut, and only 4 also fail the model-side cut. Direction agreement is 82% on all 112 points, 85% under a model-side cut alone, and 89% under the two-sided cut. The filter also removes loratadine, the only drug in the

Table 2: Blinova et al. (2018) rank agreement, multi-channel model against the hERG-only model. Correlations are Spearman ρ . The pooled 112-point row is pseudoreplicated ($28 \text{ drugs} \times 4 \text{ doses}$); the per-dose rows ($n = 28$ each) and the per-drug rows remove that dependence. “Corrected” applies the Fridericia correction to the model APD90, matching the corrected FPDc the measurement reports.

Set	Multi-channel	hERG static	hERG dynamic
All 112 points, uncorrected	0.6714	0.6180	0.5728
All 112 points, corrected	0.7023	0.6179	0.5727
Beat-only points, corrected	0.6708	0.5437	0.5200
Per-dose $1\times$, corrected	0.7760	0.6873	0.6785
Per-dose $2\times$, corrected	0.8368	0.7398	0.6835
Per-dose $3\times$, corrected	0.7739	0.6652	0.6213
Per-dose $4\times$, corrected	0.5230	0.4136	0.2294
Per-drug mean, corrected	0.7593	0.6684	0.5233
Per-drug maximum, corrected	0.7838	0.7508	0.6237

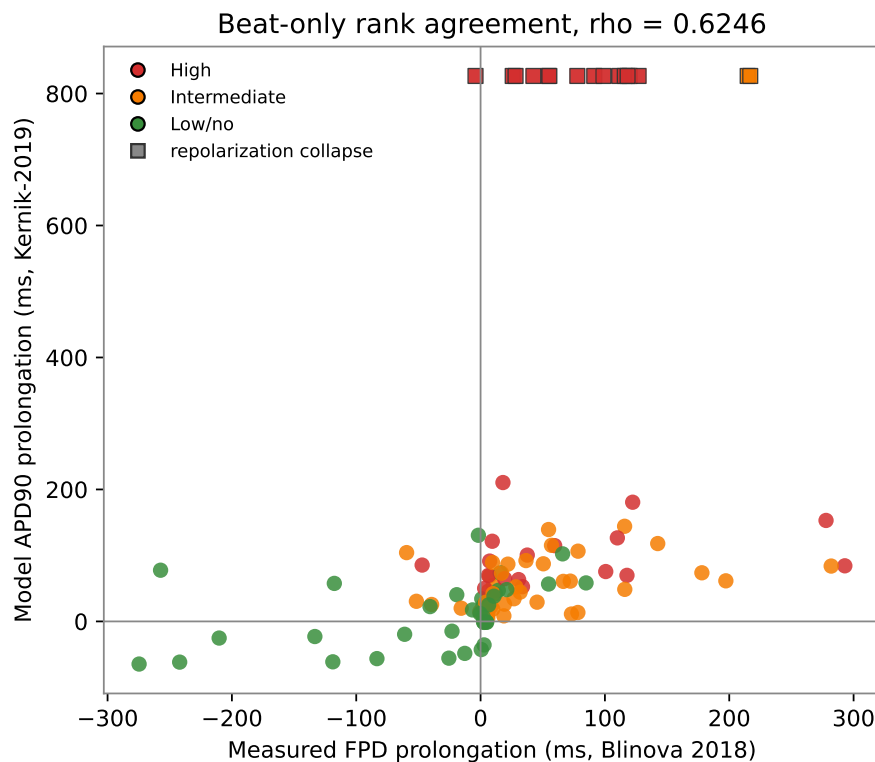


Figure 1: Model APD90 prolongation against measured FPD prolongation, 112 matched drug-dose points from Blinova et al. (2018). Colour is the CiPA TdP class. Squares are model repolarization collapses, plotted at the fixed ceiling; their vertical position is a convention, so the caption shows the beat-only correlation that excludes them. Values here are uncorrected; the rate-corrected correlations are in Table 2.

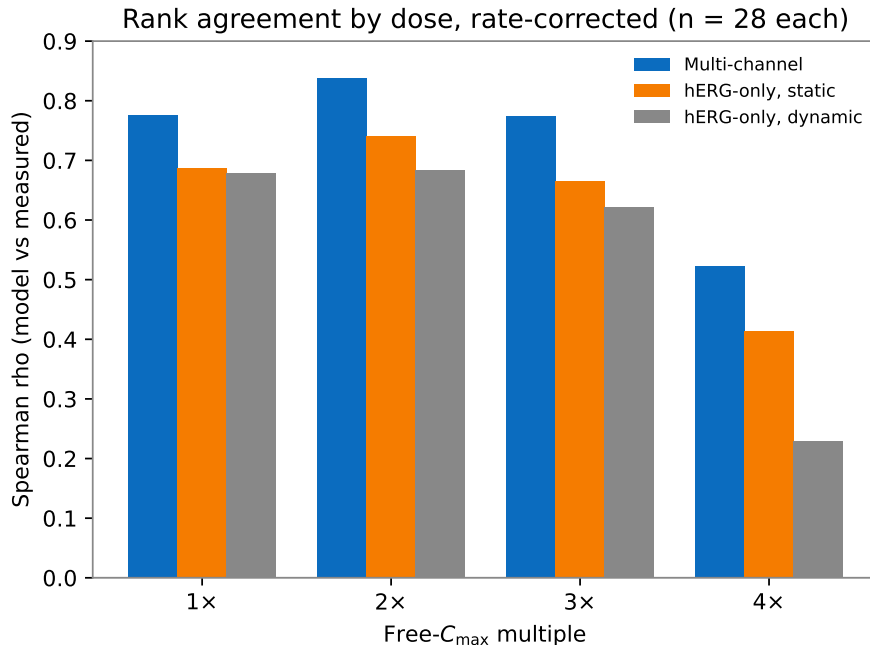


Figure 2: Rank agreement by free- $C_{\max}$ multiple, rate-corrected, for the multi-channel model and the two hERG-only models. Each bar is a Spearman correlation over the 28 drugs at that dose, so no drug is repeated within a bar. The multi-channel model leads at every dose.

panel for which Blinova annotated no EAD. We report the two-sided figure as primary and the others as sensitivity.

3.3. The Lee et al. (2025) direction check

Lee et al. measured FPDc across the same 28 CiPA drugs and published it as a signed heatmap in Figure 3 of that paper. We digitize the heatmap into an ordinal and rank the model against it. Pooled over the 75 non-quiescent drug-dose cells, the model ranks against the measured FPDc at 0.67, the same value it reaches on the Blinova panel. The pooled number hides a strong dose dependence. At $0.1\times$ and $1\times C_{\max}$ the measured ordinals are coarse and mostly tied (27 of 28 values fall in one bin at $0.1\times$), and the model ranks at 0.07 and 0.55. At $10\times$ the rank is 0.84 over 17 cells, and at $100\times$ it is 0.95 over only 4. One point per drug at its maximum measured response gives 0.75 over 28 drugs. The model recovers the measured ranking at supra-clinical exposure; at clinical exposure it only partly does.

The per-dose ranks answer the question the Introduction sets. The model does not recover the measured ranking at clinical exposure, and it does recover it as exposure rises.

We also keep a narrower direction check on ten anchored drugs (Table 4), because it isolates the calcium-blocker direction. The multi-channel model agrees on 9 of 10. The hERG-only model agrees on 6 of 10. It cannot shorten, so it misses all four calcium-channel blockers, nifedipine, nitrendipine, diltiazem, and verapamil. The multi-channel model misses only verapamil.

All eight of Lee’s high-risk drugs prolonged FPDc by more than 20% at the CiPA reference doses, and the model prolongs all eight. Ranolazine also prolongs in both; at the reference

Table 3: Model against the Lee et al. (2025) FPDc heatmap at each $C_{\max}$ ratio. Rho is Spearman over the non-quiescent cells at that ratio. Direction is the sign match; the model side uses a 5 ms deadband, the measured side is the sign of the ordinal. Ties is the number of measured values that share a bin with another cell.

$C_{\max}$ ratio	n	Spearman ρ	Direction	Tied measured cells
$0.1\times$	28	0.07	11/28	27
$1\times$	26	0.55	17/26	25
$10\times$	17	0.84	16/17	17
$100\times$	4	0.95	3/4	2

Table 4: Direction agreement on ten anchored drugs from Lee et al. (2025). TdP class is the CiPA assignment. The multi-channel column is the model of this paper; the hERG column is the hERG-only model.

Drug	TdP class	Lee measured	Multi-channel	hERG-only model
quinidine	High	prolong	prolong	prolong
dofetilide	High	prolong	prolong	prolong
ibutilide	High	prolong	prolong	prolong
sotalol	High	prolong	prolong	prolong
droperidol	Intermediate	prolong	prolong	prolong
ranolazine	Low/no	prolong	prolong	prolong
nifedipine	Low/no	shorten	shorten	prolong
nitrendipine	Low/no	shorten	shorten	prolong
diltiazem	Low/no	shorten	shorten	prolong
verapamil	Low/no	shorten	prolong	prolong

doses it is the only low/no-risk drug that Lee found to prolong FPDc past 20%, though at $10\times C_{\max}$ mexiletine joins it. The calcium blockers nifedipine, nitrendipine, and diltiazem shorten in both. The added channels are what carry that shortening direction.

3.4. Verapamil is the largest error in the dataset

Verapamil is the single direction miss in the Lee check and it is the largest absolute error in the Blinova comparison. In Blinova, measured ΔFPDc at $2\times$, $3\times$, and $4\times$ is -19 , -118 , and -257 ms. The model gives $+40$, $+58$, and $+78$ ms. That is a 335 ms error at the top dose, and it is larger than the error on any other drug. Verapamil is a calcium blocker whose hERG block is small, so the model’s hERG-dominated prolongation exceeds its calcium shortening. This is the known calcium-block limitation of a hERG-weighted read, and it is the model’s worst case. The Lee check reproduces it, so two measured datasets show the error. The free- $C_{\max}$ sweep also moves verapamil, but that is a rerun of the model and not independent evidence.

3.5. The collapse flag carries almost no information

The collapse flag looks better in raw counts than it is. The model collapses on six drugs, and Blinova annotated an EAD for all six, so no collapse was a false positive. But Blinova annotates an EAD somewhere for 27 of the 28 drugs, so the negative class holds a single drug, loratadine. The full 2×2 is 6 collapse-with-EAD, 0 collapse-without-EAD, 21 no-collapse-with-EAD, and 1 no-collapse-without-EAD. A Fisher exact test on that table gives $p = 1.0$. Drawing six drugs at random from a panel with 27 EAD drugs puts all six in the EAD class 79% of the time. The claim that no collapse was a false positive restates the base rate. Sensitivity is the only informative count, and it is poor: the flag caught 6 of the 27 EAD-bearing drugs (22%).

The collapse flag is more useful as a severity score than as a binary. Collapse severity, the model’s fractional prolongation short of the ceiling, ranks the measured EAD rate at Spearman $\rho = 0.72$ ($p = 1.5 \times 10^{-5}$) and separates EAD-bearing from EAD-free drugs at AUC 0.81. That is a stronger result than the binary flag and belongs in the paper as a quantitative readout, with the caveat that the EAD-free class is a single drug.

4. Negative and null results

The population confidence signal is overconfident under both population models. A population-of-models run (10 conductance draws, frozen cuts) assigns each drug a confidence equal to the fraction of cells agreeing on the risk class. That confidence is not calibrated (Fig. 3). Under independent draws the point-biserial correlation between confidence and correctness is +0.17 ($p \approx 0.38$ at $n = 28$), and the top band reports a mean confidence of 0.99 while its accuracy is 0.63 ($n = 19$). Under correlated draws the correlation is -0.06 , and the top band (mean confidence 0.99, $n = 25$) has accuracy 0.56. We report both modes rather than the positive one alone. The population argmax does not beat the deterministic call (0.57 against 0.61). The confidently-wrong drugs are the low-risk over-call set plus sotalol, the same structural bias as the benchmark. We do not publish this confidence as a probability.

Adding I_{Ks} is null. Adding the slow delayed rectifier I_{Ks} to the risk score leaves the 28-drug accuracy unchanged (17 of 28) and moves no drug’s class; the largest movement is 0.001 in fractional APD90. The result is null, but its scope is narrow: a reference I_{Ks} IC50 exists for only 7 of the 28 drugs, so the other 21 are unchanged by construction. The reference I_{Ks} IC50s block too little at 1–4 $\times$ free C_{max} to move this score. This is not a claim that I_{Ks} is irrelevant to repolarization.

Adding I_{NaL} is negative, and the churn is large. Adding an O’Hara-Rudy-style late sodium current lowers accuracy from 17 of 28 to 15 of 28, and from 9 of 16 to 7 of 16 on the blind split. The effect is not one correction. Nine of the 28 classes change. Three move to their true class (mexiletine to Low, sotalol and disopyramide to High), and six regress (verapamil to High and collapse, and cisapride, clarithromycin, droperidol, pimozide, and terfenadine over-called High). High-risk false positives go from 1 to 4. The cause is a shift in the baseline: at the pre-specified conductance ($g_{NaL} = 0.10$ mS/ μ F, a 1.03% sustained late fraction) the drug-free baseline APD90 lengthens from 413 ms to 588 ms. A 1% persistent current should

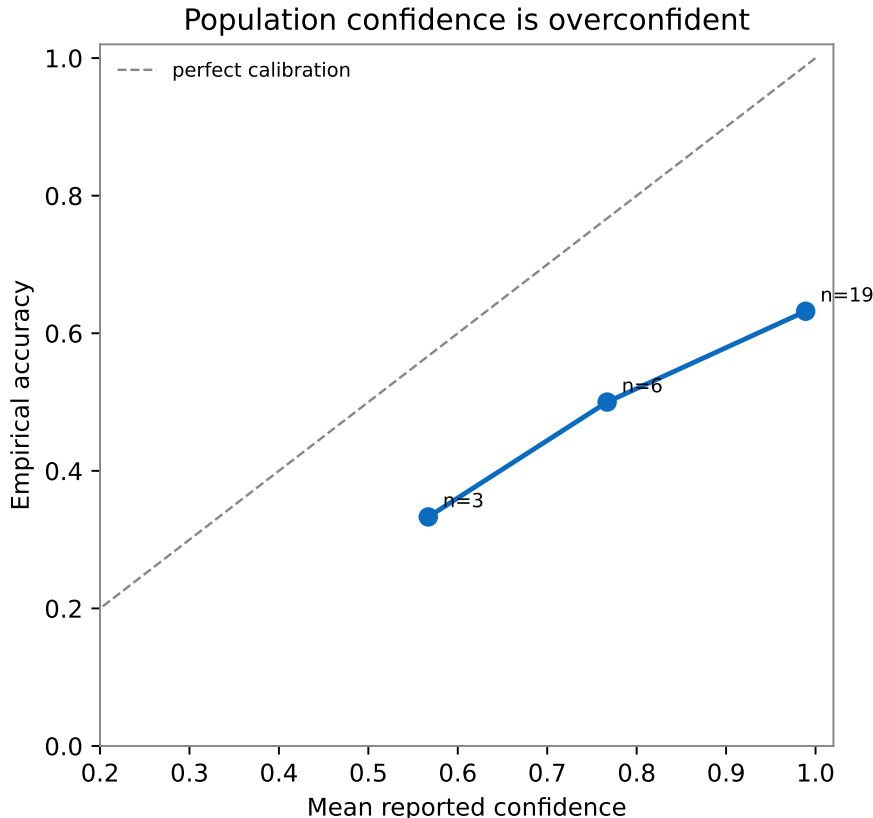


Figure 3: Reliability bands for the population confidence signal, independent-draw mode. Points fall below the identity line: reported confidence overstates achieved accuracy. The correlated-draw mode is flatter still ($\rho = -0.06$) and is reported in the text.

not extend repolarization by 42%, so either the conductance is large for this model or the model holds thin repolarization reserve. The accuracy drop may follow from that baseline shift rather than from the late current itself. The negative result is therefore conditional on g_{NaL} , not a property of I_{NaL} in general.

The model’s mechanism recovery fails. A main use of the model is a mechanism ranking: which currents a compound blocks. We tested whether the model recovers the true current profile from its own output, over a grid of single-current and current-pair hypotheses. Single-current recovery succeeds on 4 of 8 test cases. Pair recovery succeeds on 2 of 8, and the two successes are both verapamil, the one drug whose pair the grid resolves. This is a self-test against targets the same model generated, so it is not validation against wet data, but it is a failure of the internal consistency the product depends on, and we report it because we declare a competing interest in the model.

The exposure input is a first-class uncertainty. The risk call hangs on the supplied free C_{max} . Sweeping it over $0.5\text{--}2.0\times$ with the cuts frozen leaves 17 of 28 calls stable and moves 11. High-risk sensitivity rises from 4 of 8 to 7 of 8 across the range, and high-risk false positives from 0 to 6. Six of the 11 flips are the model hitting the collapse ceiling. A read that quotes a class without quoting the exposure it assumed is under-specified.

5. Discussion

The three-class CiPA risk label is decided against the multi-channel model. A hERG-only model, fit and frozen on the same split, scores 18 of 28 on the static hERG input and 19 of 28 on the dynamic input, against 17 of 28 for the model of this paper. The label is a threshold on a hERG-dominated risk, so a current that does not prolong the action potential on its own rarely changes the class. Adding I_{CaL} and I_{Na} block moves the repolarization duration without changing the class. On this endpoint the extra channels add mechanism without adding accuracy.

The added channels improve the continuous readout. The multi-channel model reaches a rank correlation of 0.70 against Blinova et al., against 0.62 for the hERG-only model, and it agrees with the measured direction on 89% of the measurable points against 78% ($p = 0.005$). The hERG-only model cannot shorten, so it scores the always-prolong base rate and misses every calcium-channel blocker. The multi-channel model recovers the shortening, and that recovery is the source of its whole advantage. The channels do what they were added to do, on the readout where their mechanism is visible.

The result sets the test for future claims. A multi-channel model offered as a better classifier has to beat the hERG-only model on the three-class label, on the same split and protocol. This model does not. A multi-channel model offered as a mechanistic or continuous readout should be judged on that readout. Both claims can hold at once, and a paper that reports only the second can appear stronger than it is. In short, the extra channels improve the continuous readout. The regulatory risk class is unchanged.

6. Limits of this validation

Established: the multi-channel model reproduces the rank and direction of measured repolarization prolongation better than a hERG-only model. The added channels carry the calcium-block shortening direction that the hERG-only model cannot produce.

Not established:

- Any advantage of the multi-channel model on the three-class CiPA risk label, where the hERG-only model is equal or better.
- Waveform or magnitude equivalence.
- A rank correlation over the image-only supplement tables of Lee et al. We did not digitize those values, so the full-panel rank we report rests on the coarse Figure 3 bins.
- The mechanism ranking that is the main use, which has been tested only against phenotypes the same model generated.

Four further limits apply. The model is a single deterministic myocyte, so a model response is a point while a measured response is a distribution; the population run addresses this only for the confidence signal. Both datasets are MEA field-potential recordings, so neither tests the model against a patch-clamp action potential. The Fridericia correction aligns the model and the measurement, but the model's APD90 and the measured FPD remain different quantities. The 5 ms filter is defined on the measurement and removes the one drug without a measured EAD.

Data and code

The committed data, results, and analysis scripts are under `cipa_validation/`. The Blinova comparison is produced by `blinova_validation.py`, the comparison against the hERG-only model by `margin_comparison.py`, and the rate correction by `rate_correction.py`. The Lee panel is digitized by `lee_fig3_extract.py`, checked by `lee_fig3_verify.py`, and ranked by `lee_comparison.py`. The calibration analysis is `i2_calibration.py`; the I_{Ks} , I_{NaL} , collapse-severity, and mechanism-recovery variants are `cipa_validation_harness_iks.py`, `cipa_validation_harness_inal.py`, `collapse_calibration.py`, and `i1_mechanism_recovery.py`. The figures are regenerated from the committed result files by `make_preprint_figures.py`. A versioned copy of the scripts and the result files is available at <https://github.com/Perturb-Bio-Inc/cipa-validation> (tag v1.0.0) and archived at OSF, DOI 10.17605/OSF.IO/JV4DZ.

Declarations

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Competing interests. The author is the sole founder of Perturb Bio, which developed the in-silico read evaluated in this paper.

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